

Prabhu Pugalenth

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EDUCATION

University of Southern California Master of Science in Computer Science Relevant Courses: Applied Natural Language Processing, Deep Learning and its Applications, Statistical Machine Learning Focus Areas: Machine Learning Systems, Deep Learning Architectures, Generative AI, Representation Learning	Los Angeles, CA Jan 2026 - Expected May 2027
Coimbatore Institute of Technology Master of Science in Software Systems Relevant Courses: Advanced Data Structures and Algorithms, Artificial Intelligence, Operating Systems, Computer Networks	Tamil Nadu, India Sep 2020 - May 2025

EXPERIENCE

University of Southern California Graduate Research Assistant	Los Angeles, CA Feb 2026 – Present
- Extending CogRouter, an ACT-R-inspired cognitive routing framework, to generalize across agentic and mathematical reasoning domains - Designed and implemented a training-free online token ledger that monitors per-episode budget consumption and dynamically re-plans cognitive depth allocation mid-episode - Built a confidence-triggered verification gate for math reasoning, enforcing self-refinement when answer confidence falls below adaptive thresholds on MATH500, AIME24, and GSM8K - Developing a cost-conditioned cognitive router retrained via GRPO with per-tool API cost signals to reduce token expenditure on expensive tool calls in web and agentic environments - Evaluating framework generality across six benchmarks spanning two task domains: agentic (WebArena, GAIA, AssistantBench) and mathematical reasoning (MATH500, AIME24, GSM8K)	
Ernst and Young Data Analyst Intern	Tamil Nadu, India Dec 2024 – May 2025
- Built Python and SQL pipelines to ingest, clean, and analyze large enterprise datasets - Developed analytics dashboards (Power BI) to monitor operational metrics and trends - Implemented data validation and monitoring to ensure reliability of reporting workflows - Collaborated with cross-functional teams to troubleshoot data issues and optimize queries - Automated data workflows using CI/CD pipelines for validation and reproducible reporting - Documented data models, query logic, and reporting pipelines for maintainable analytics systems	
Ernst and Young Software Developer Intern	Tamil Nadu, India Jun 2023 – Dec 2023
- Developed Python-based backend services to process and extract structured insights from large document datasets - Designed and deployed REST APIs using Docker containers with CI/CD pipelines for automated testing and integration - Optimized model inference pipelines and implemented logging and monitoring for reliability - Worked in collaborative engineering workflows using Git version control and code review practices	

RESEARCH

Reasoning Economics Environment for LLMs RL, LLM Systems, GRPO, TRL	Jan 2026 - Apr 2026
- Designed a reinforcement learning environment to study compute-optimal reasoning by training LLMs to allocate tokens adaptively across queries of varying difficulty under shared budget constraints - Formulated the problem as a knapsack-constrained MDP with states encoding query complexity, remaining token budget, and sequential task dependencies - Implemented GRPO-based training using TRL's GRPOTrainer, validating the pipeline end-to-end on Qwen3-1.7B and scaling experiments to Qwen3-8B on Lambda Labs A100 instances - Built reward functions and evaluation pipelines measuring episode accuracy, budget utilization, and reward trajectories across difficulty-stratified datasets - Benchmarked learned policies against uniform, greedy, and oracle baselines; resolved vLLM/transformers dependency conflicts and CUDA OOM errors across HPC and cloud environments	
LLM-Based Social Survey Simulation NLP, LoRA Fine-tuning, Statistical Evaluation	Feb 2026 - Apr 2026
Developed LLM-based frameworks to simulate population-level survey responses from limited pilot samples, targeting data-scarce settings where full survey deployment is cost-prohibitive	

- Fine-tuned Qwen2.5-7B-Instruct and Qwen3-8B using LoRA (via TRL/PEFT) across split configurations; best results achieved at 50/10/40 train/val/test split over 3 epochs
- Designed evaluation metrics including item-level Pearson correlation, Earth Mover's Distance (EMD), and Mean Absolute Error (MAE) to measure distributional and structural fidelity
- Implemented pilot-conditioned few-shot prompting to inject base-rate priors and reduce systematic bias in LLM-generated survey data
- Built reproducible pipelines for large-scale simulation, statistical comparison against real datasets, and analysis of accuracy-distribution alignment trade-offs under data-scarce settings

CogRouter: LLM Safety & Trustworthiness Benchmarking | VLMs, Agent Evaluation Apr 2026 - Present

- Conducting research under Prof. Emilio Ferrara (USC ISI) to evaluate open-source vision-language models on AI safety dimensions including jailbreak resistance, bias, and agent-level harmful behavior
- Integrated Qwen3-VL-8B-Instruct into the OSWorld benchmark by building a custom **Qwen3VLAgent** class with tool-call action parsing, smart-resize image preprocessing, coordinate scaling, and history-aware multi-turn context management
- Deployed evaluation infrastructure across USC CARC (Endeavour, Apptainer/Slurm) and Lambda Labs A100 instances; served the model via vLLM's OpenAI-compatible endpoint (~33GB VRAM utilization)
- Extended benchmarking to the OS-HARM safety suite to measure attack success rate (ASR) and task success rate across desktop agent scenarios involving potentially harmful instructions
- Patched OSWorld's **agent.py** routing layer to support local vLLM-served models and contributed agent integration code enabling reproducible safety evaluation of new open-source VLMs

PROJECTS

Forecasting of Postprandial Blood Glucose | Time-Series ML, Healthcare AI Sept 2025 - Dec 2025

- Developed diffusion-based deep learning models for multivariate physiological time-series forecasting using CGM data and contextual signals (meals, insulin, activity)
- Designed preprocessing pipelines to align irregular clinical data, handle missing values, and encode temporal dependencies
- Trained and optimized models using RMSE/MAE-based objectives, achieving accurate prediction of post-meal glucose dynamics over a 200-minute horizon
- Evaluated generalization across unseen individuals and analyzed temporal patterns such as peak timing and recovery behavior
- Investigated robustness under distribution shifts and validated clinical plausibility using statistical and domain-specific metrics

DRILL: Competitive Outcome Prediction | XGBoost, CatBoost, Large-Scale ML Aug 2025 - Dec 2025

- Modeled round-level win probability in CS:GO using large-scale event-based gameplay data
- Engineered high-dimensional features capturing game state (economy, player health, utility usage, positional context)
- Trained gradient-boosted models (XGBoost, CatBoost) achieving strong predictive performance (ROC-AUC ~ 0.87)
- Evaluated probabilistic calibration using LogLoss, Brier score, and Expected Calibration Error (ECE)
- Analyzed model performance across match scenarios to identify failure modes under distributional shifts

Intelligent Log Analysis & Incident Triage System | Python, ML, Systems, Backend Apr 2025 - Sept 2025

- Designed and implemented a modular log analysis system to ingest, normalize, and process large-scale application and system logs in Linux environments
- Developed ML-assisted and rule-based methods to detect anomalies, recurring error patterns, and incident signatures across multiple log sources
- Evaluated system performance and robustness across diverse log datasets to ensure reliability and consistency
- Exposed analysis results via REST APIs and automated triage outputs to support debugging workflows and system reliability analysis

SKILLS

Programming: Python, C/C++, SQL, Bash

Machine Learning: Deep Learning, Reinforcement Learning, LLM Systems, Model Evaluation

Research: Experimental design, benchmarking, statistical evaluation, distribution analysis

Frameworks: PyTorch, Scikit-learn, NumPy, Pandas, TRL, PEFT, vLLM

Systems: ML pipelines, REST APIs, Linux/HPC environments, Docker, Apptainer, CI/CD workflows

Tools: Git/GitHub, Power BI, Jupyter, Slurm